



Marine Biology

海洋生物学



Yanan Di (邸雅楠)
diyanan@zju.edu.cn

Institute of Marine Biology and Pharmacology
Ocean College, Zhejiang University

Scientific Method

- A systematic way of testing ideas
- The heart is the need to **OBSERVE** nature
- Begins with a **hypothesis**, or a **testable statement** constructed from observations and reasoning
- The hypothesis must **be worded** so that it is testable by scientific methods

Hypothesis: There are mermaids in the ocean!

**Hypothesis: There are mermaids appeared in E
122-123 degree and S 52-53 degree!**

Two ways of thinking

1. Induction (归纳) : General conclusions are made on the basis of specific observations



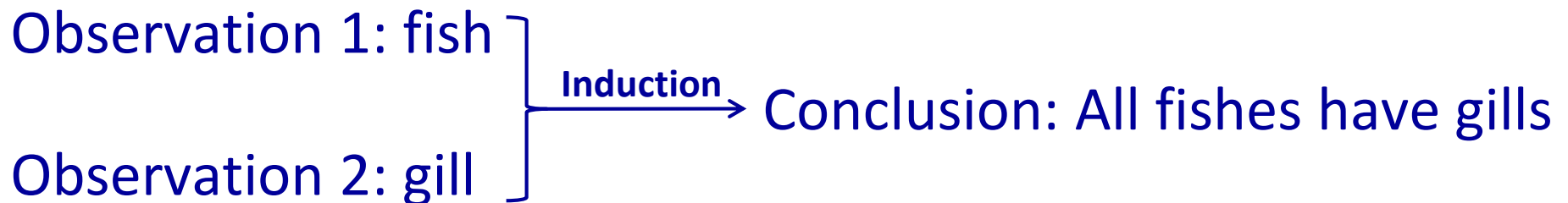
Sailfish (旗鱼)



Shark (鲨鱼)



Tuna (金枪鱼)



2. Deduction (演绎) : Specific predictions are made by applying the general principles



Observation: Marine Animal & Gills

Induction



Conclusion: All marine animals have gills

Deduction



Conclusion 1: Whales have gills

Conclusion 2: Whales are not marine animals

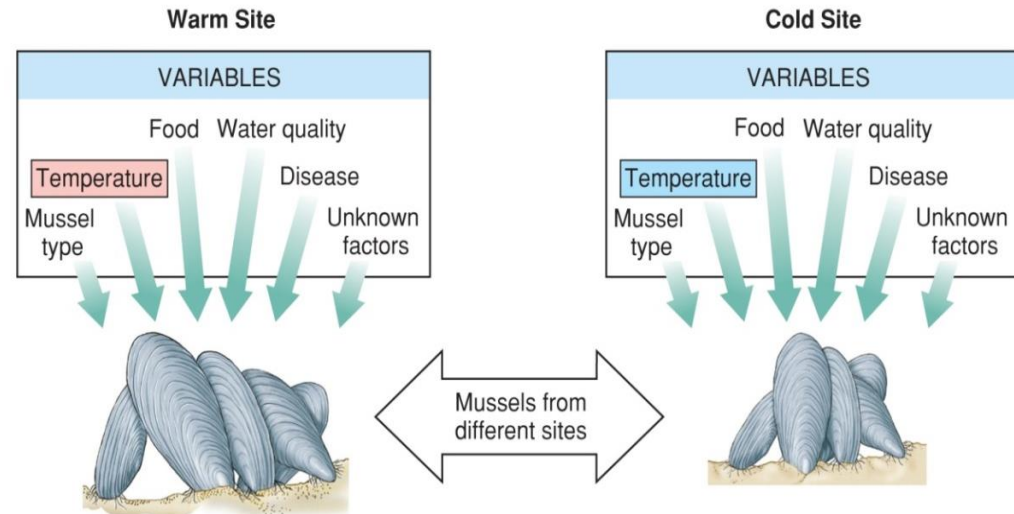
- **Variables:**

- A researcher must look at only **one factor at a time**, these factors are variables

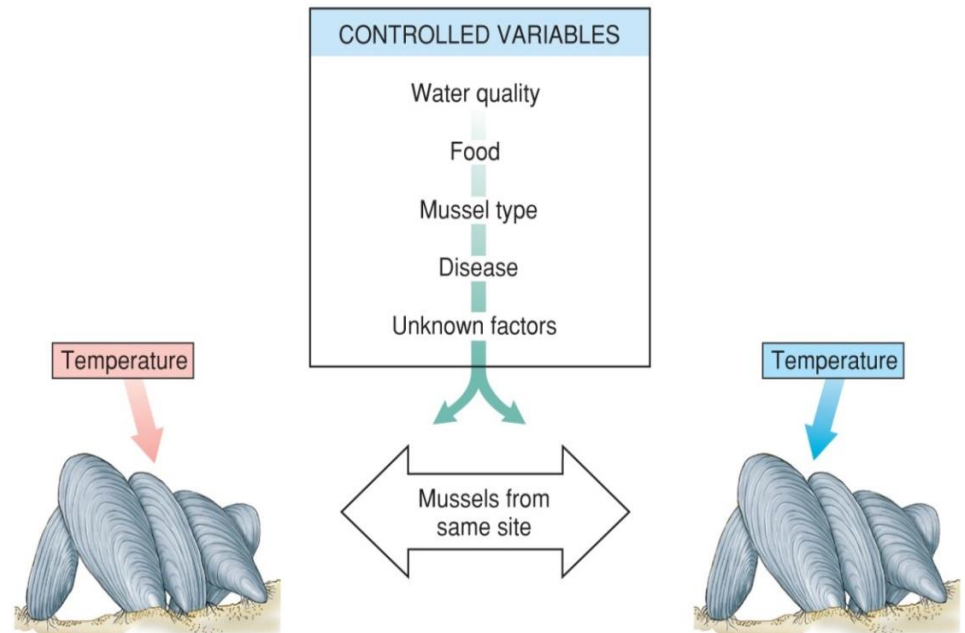
- Variable selected based on field observation

- Variables are easily controlled in lab condition than in field

Field Observations



Controlled Laboratory Experiment



- One important aspect of science and the scientific method is that you **CANNOT ABSOLUTELY PROVE** any hypothesis – you can easily **DISPROVE** hypotheses based on the evidence you have
- A hypothesis can be considered to be true based on the available evidence if it has been **tested repeatedly** and not disproven
- **Theory**
 - In daily language, it is what you suspect to be true
 - In scientific language, it is a hypothesis that has been tested repetitively over time by many people and has not been disproven
- **Truth**
 - Mountains of evidence support the theory and it is regarded as **truth** in the scientific community

- One important aspect of science and the scientific method is that you **CANNOT ABSOLUTELY PROVE** any hypothesis – you can only **DISPROVE** hypotheses based on the evidence you have
- A hypothesis can be considered to be true based on the available evidence if it has been **tested repeatedly** and not

Hypothesis —————> **Theory** —————> **Truth**

- In scientific language, it is a hypothesis that has been tested repetitively over time by many people and has not been disproven
- **Truth**
 - Mountains of evidence support the theory and it is regarded as **truth** in the scientific community

• Limitations of Scientific Method

- Due to the requirements for direct observations or measurements and a testable hypothesis, not all questions can be answered
- Science can offer no answers on values, feelings and beliefs. These are beyond the scope of the scientific method
- Scientific methods and observations can change at different time stage





The Sea Floor

The Water Planet

1. Geography of the Ocean Basins

- The ocean is the predominant feature on the Earth in total area
 - In the **Northern Hemisphere**, 61% of the total area is ocean
 - In the **Southern Hemisphere**, about 80% is ocean
- The world ocean is divided into **four/five large basins: Pacific, Atlantic, Indian, Arctic/and Southern Ocean**



(a) Southern Ocean



(b) Arctic Ocean



Table 2.1 Average Depths and Total Areas of the Four Major Ocean Basins

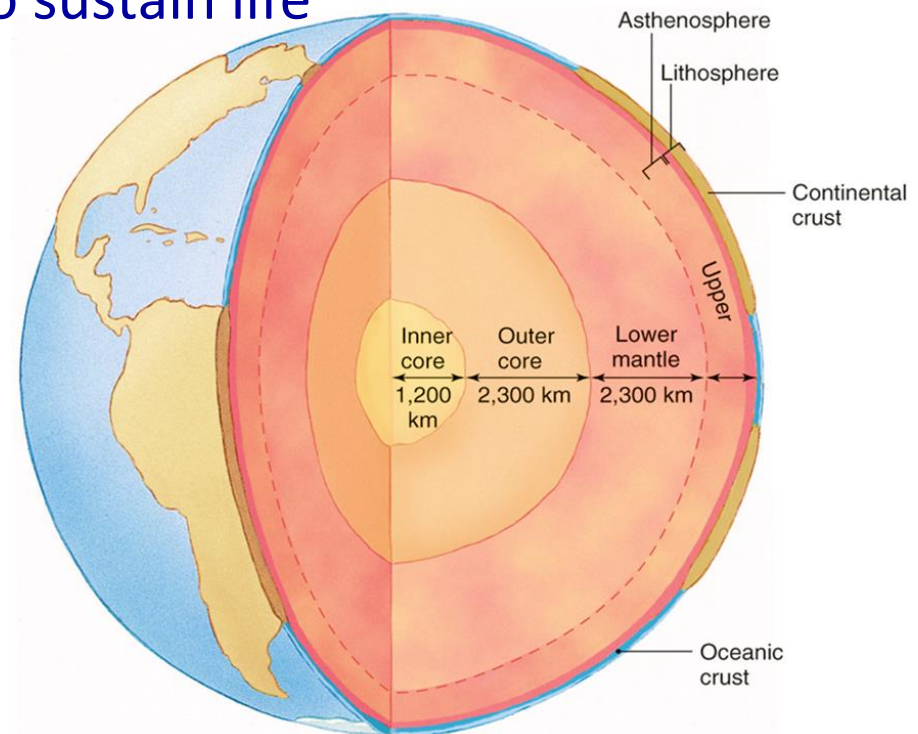
Ocean	AREA		AVERAGE DEPTH		Deepest Place
	Millions of km ²	Millions of mi ²	Meters	Feet	
Pacific	166.2	64.2	4,188	13,741	Mauriana Trench, 10,994 m (36,070 ft)
Atlantic	86.5	33.4	3,736	12,258	Puerto Rico Trench, 8,605 m (28,233 ft)
Indian	73.4	28.3	3,872	12,704	Java Trench, 7,725 m (25,344 ft)
Arctic	9.5	3.7	1,330	4,364	Molloy Deep, 5,608 m (18,400 ft)

2. Structure of the Earth

- Some evolutionary scientists believe the earth originated **4.5 billion years ago** from a cloud or clouds of dust
- The dust is the debris remaining from a great cosmic explosion——**Big Bang (宇宙大爆炸)**, which astrophysicists estimate occurred about **13.7 billion years ago**
- Earth's location relative to the sun allows for water to stay liquid – an essential element to sustain life

- Internal Structure of the Earth (**density dependent**)

1. Core: most inner layer (solid inner core and liquid outer core); iron-rich
2. Mantle: middle layer; semi-plastic composition
3. Crust: outermost layer; thinnest portion of the Earth





抖音

抖音号: 49243272619

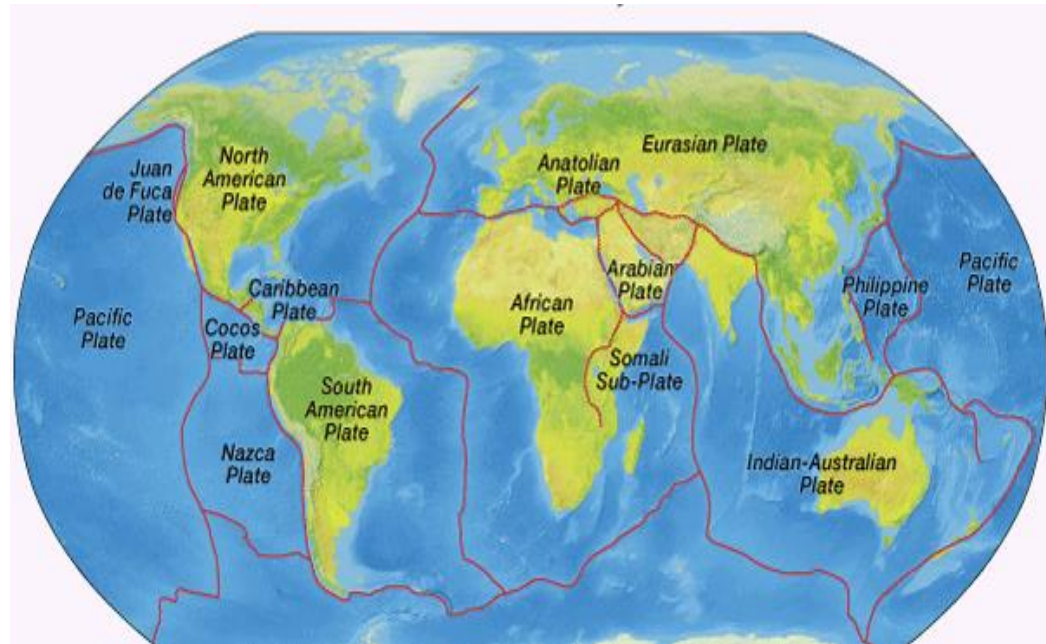
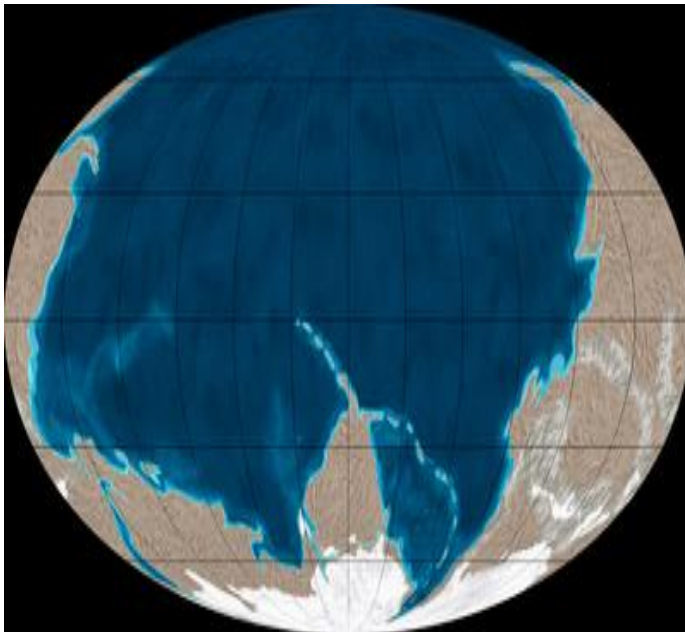


- Ocean Drilling Program (ODP,大洋钻探计划) to IODP (2003)

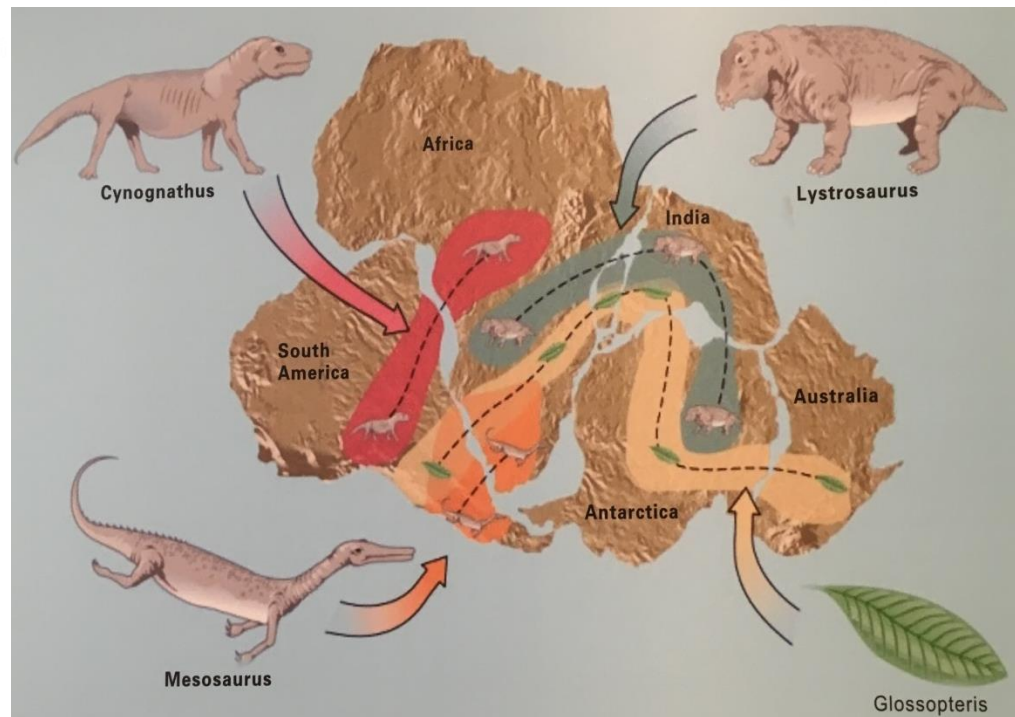
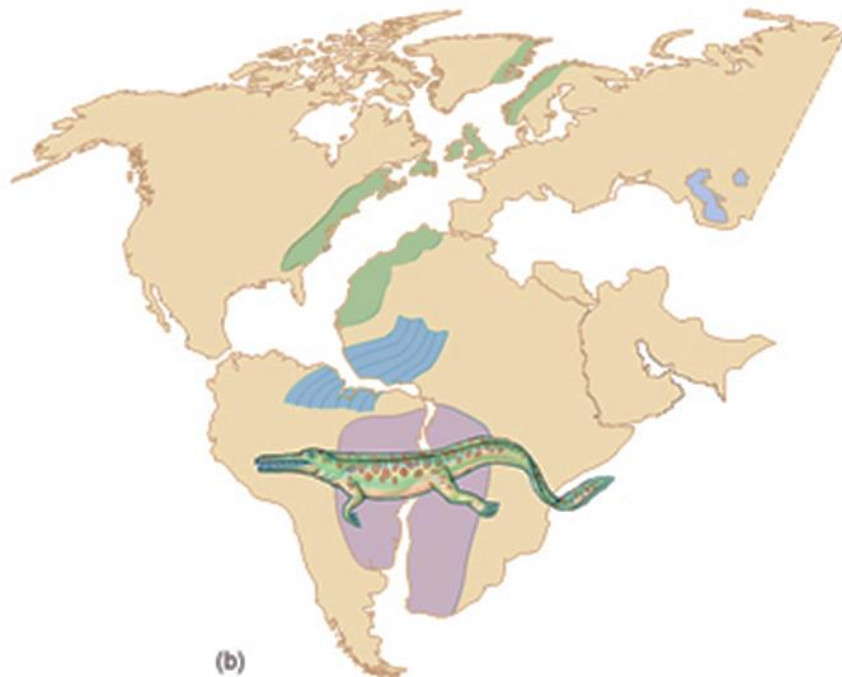
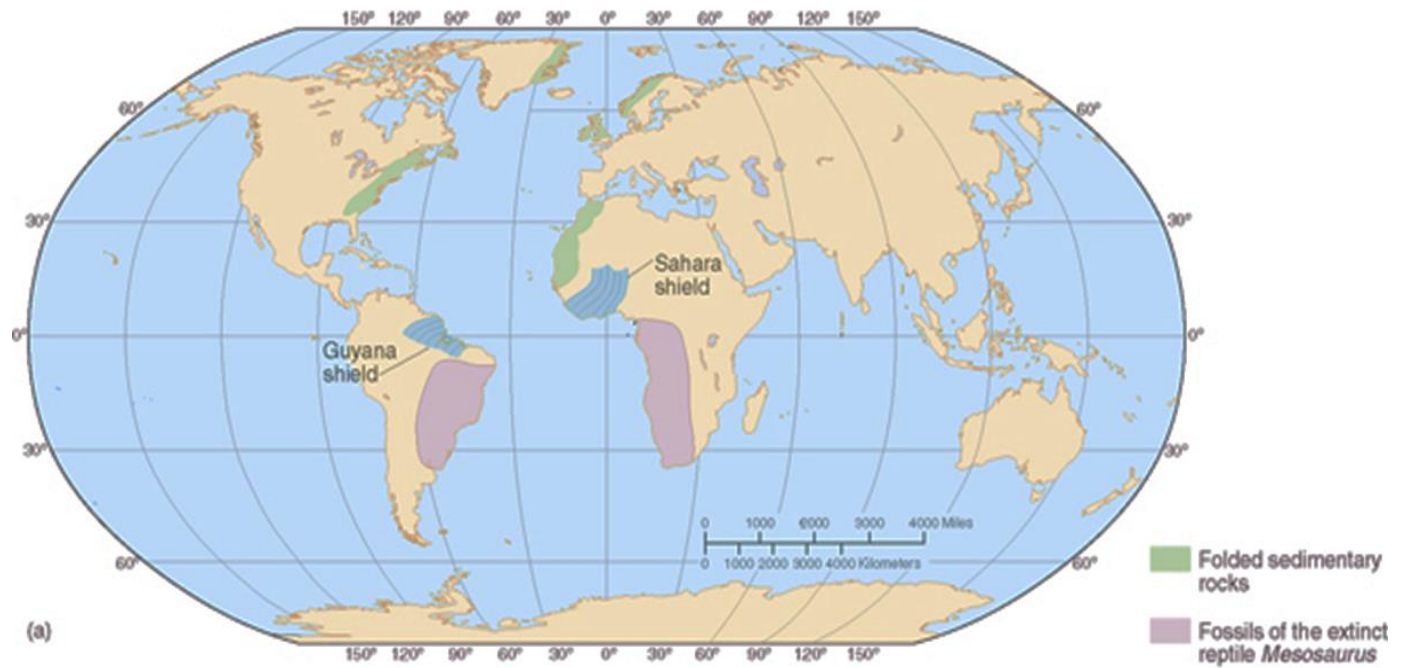
The Origin and Structure

1. Continental Drift (大陆漂移)

- Proposed in 1912 by **Alfred Wegner** (Germany)
- All continents had been joined in a single supercontinent (超级大陆) which named **Pangaea** (泛古大陆)
- Pangaea began breaking up **over 200 million years ago**
- Not widely accepted because he could not explain **HOW** this occurred

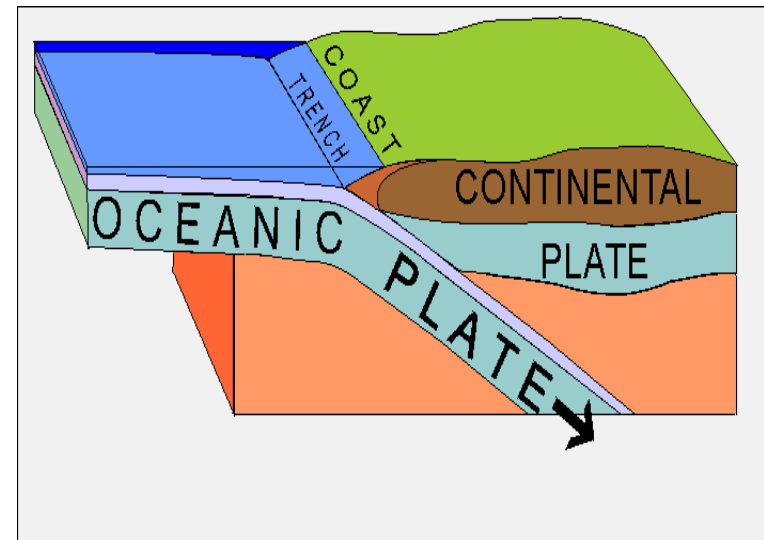
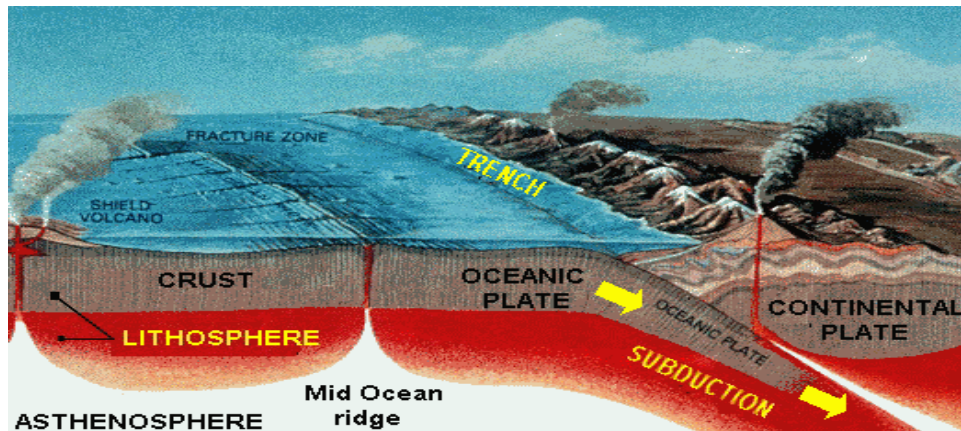


Evidence:



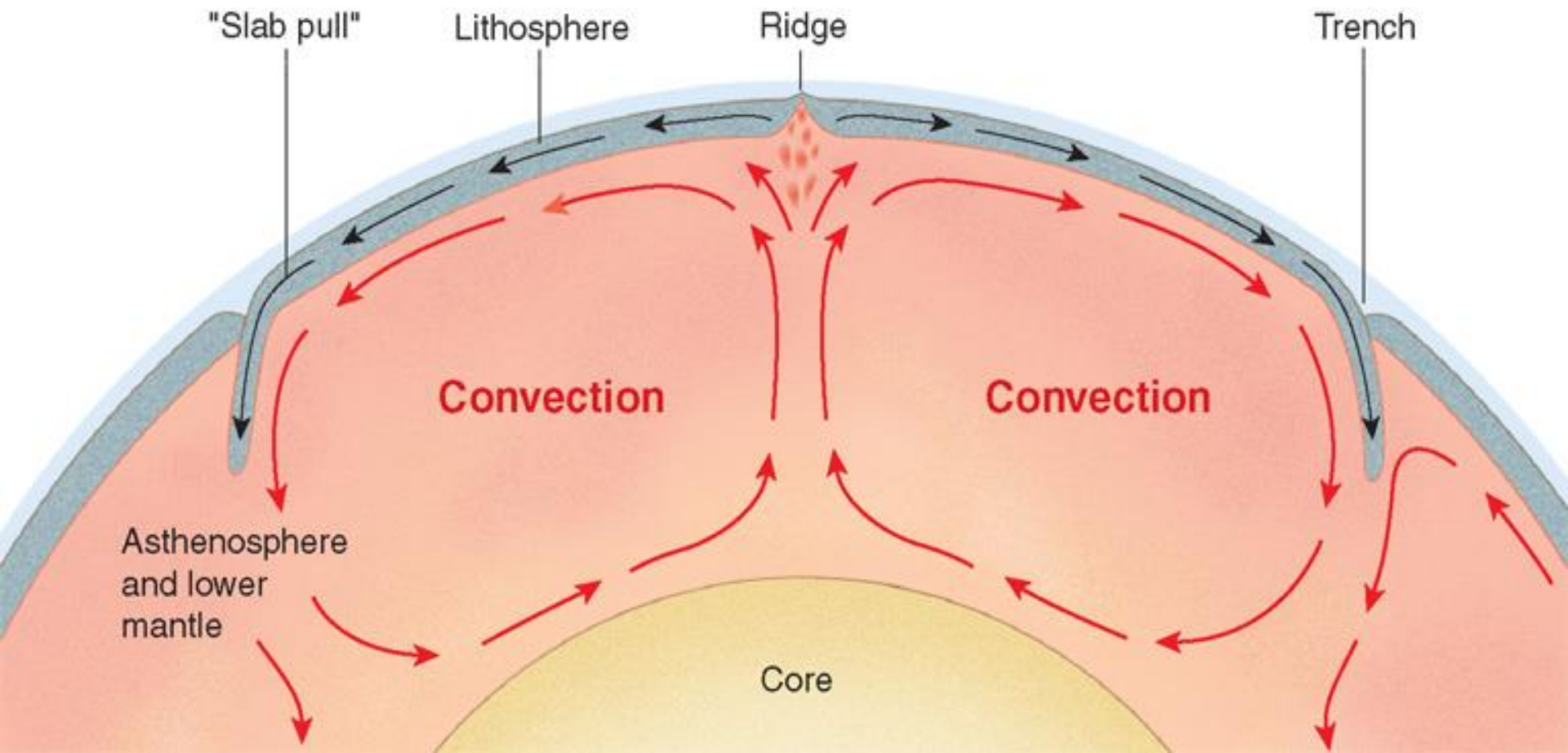
2. Plate Tectonics (板块构造)

- The main features are:
 - The Earth's surface is covered by a series of crustal plates
 - The ocean floors are constantly moving; spreading in the center and sinking at the edges and being regenerated
 - Convection(对流) currents beneath the plates assist movement
 - Heat from the mantle drives these currents
- Two primary tectonics process:
 - Seafloor spreading – creating new plate
 - Subduction(俯冲) – destroying plate



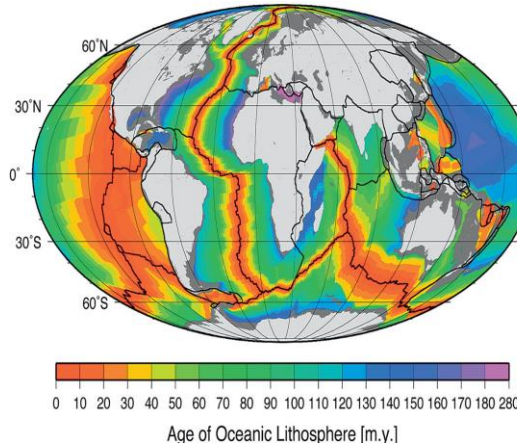
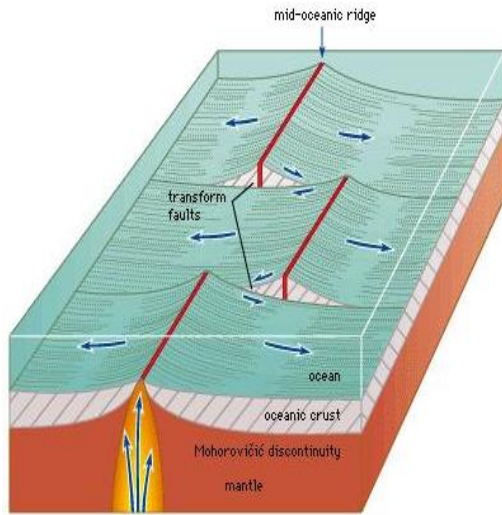
What force makes the plates move?

Heat and gravity Convection suggests that continents move on a “soup” of rock, Whereas **slab pulling** indicates that as rock cools, it sink beneath the surface and “pulls” lithosphere into the depths (*This is the currently accepted explanation)

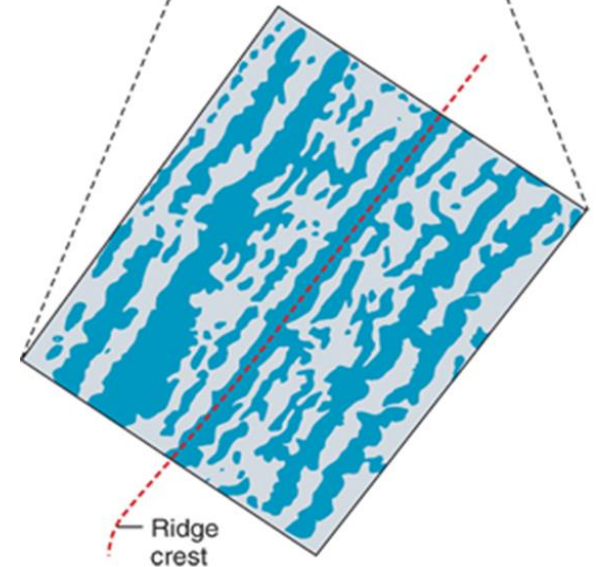
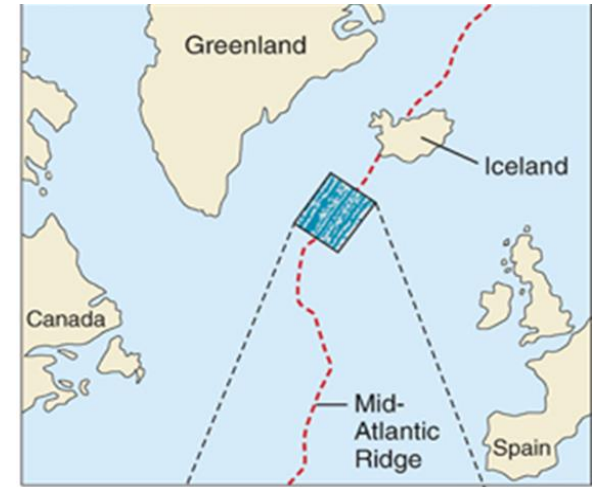


- Evidence for Plate Tectonics

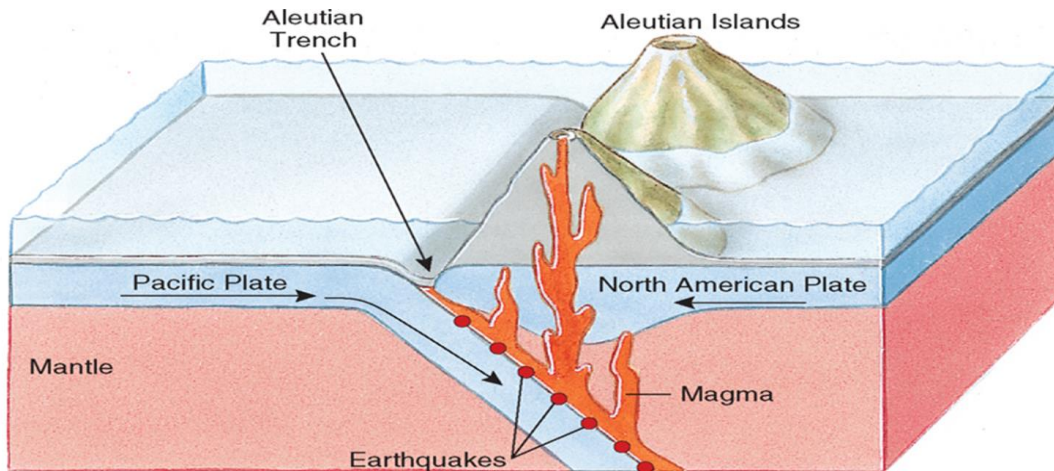
a) Mid-Oceanic Ridges (大洋中脊)



b) Geomagnetic Anomalies (地球磁场变化):



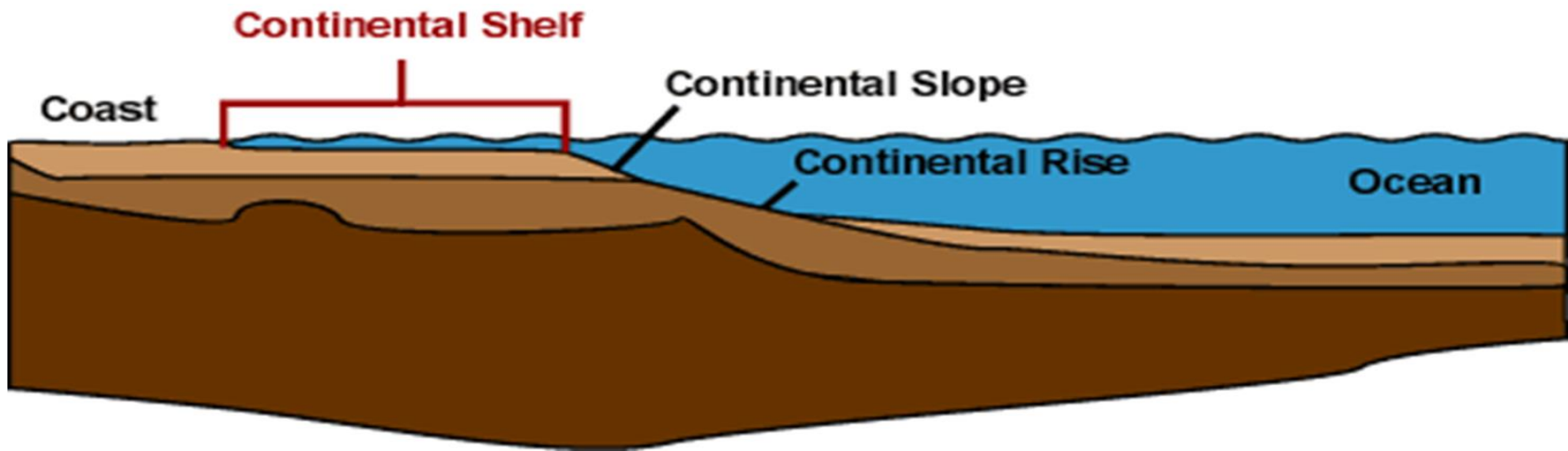
c) Island Arcs (弧形列岛)

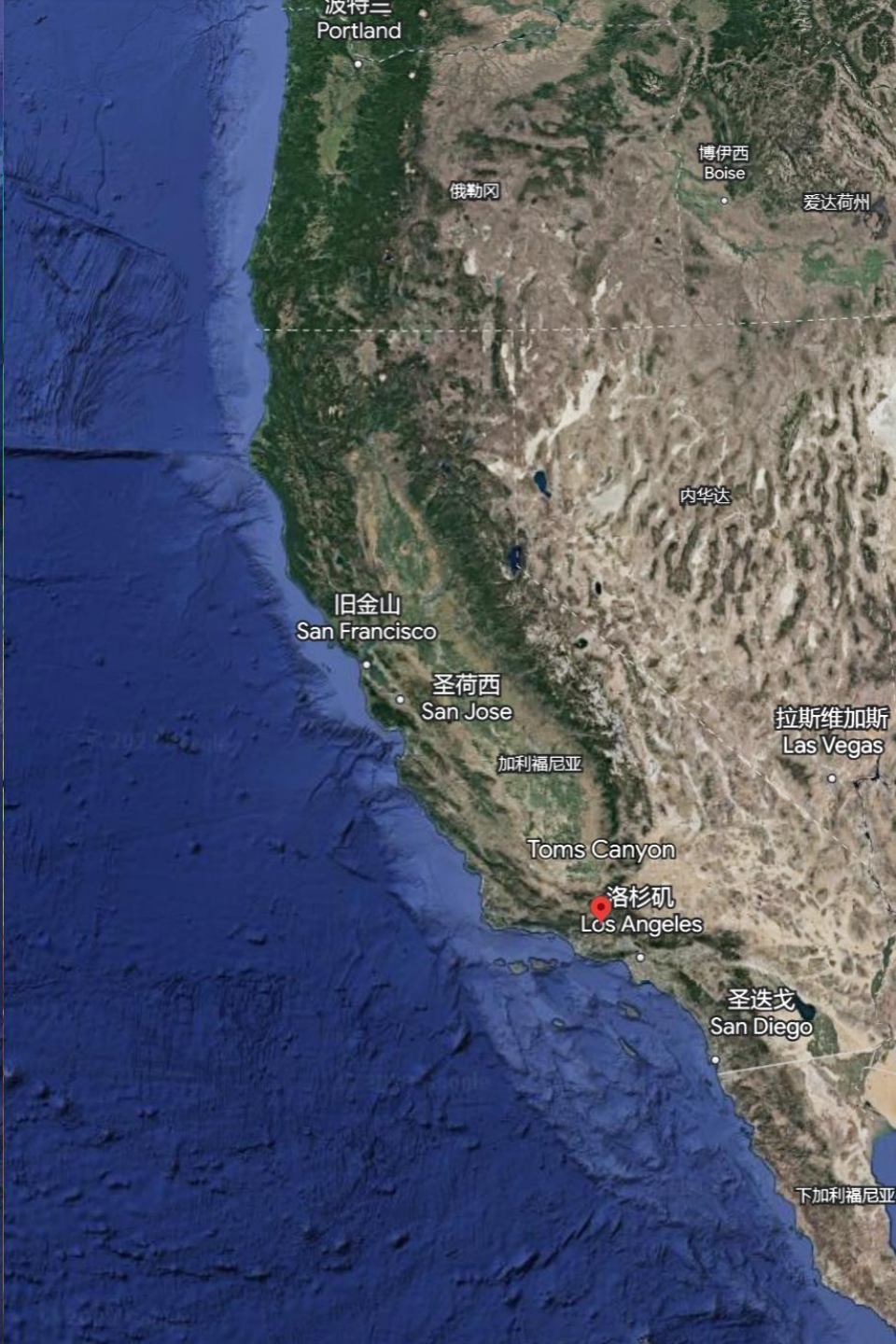
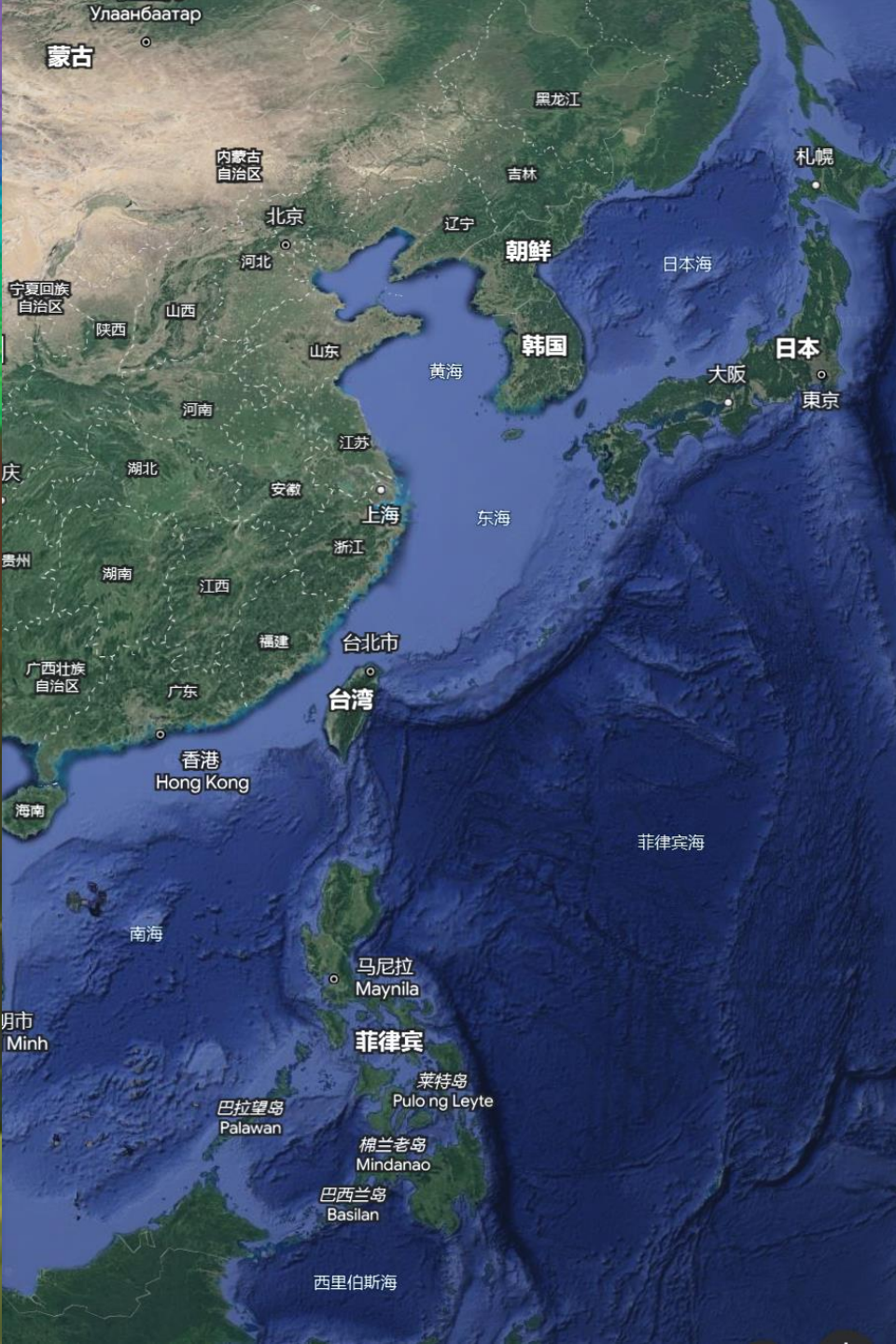


The Geological Provinces of the Ocean

1. Continental Margins (大陆边缘)

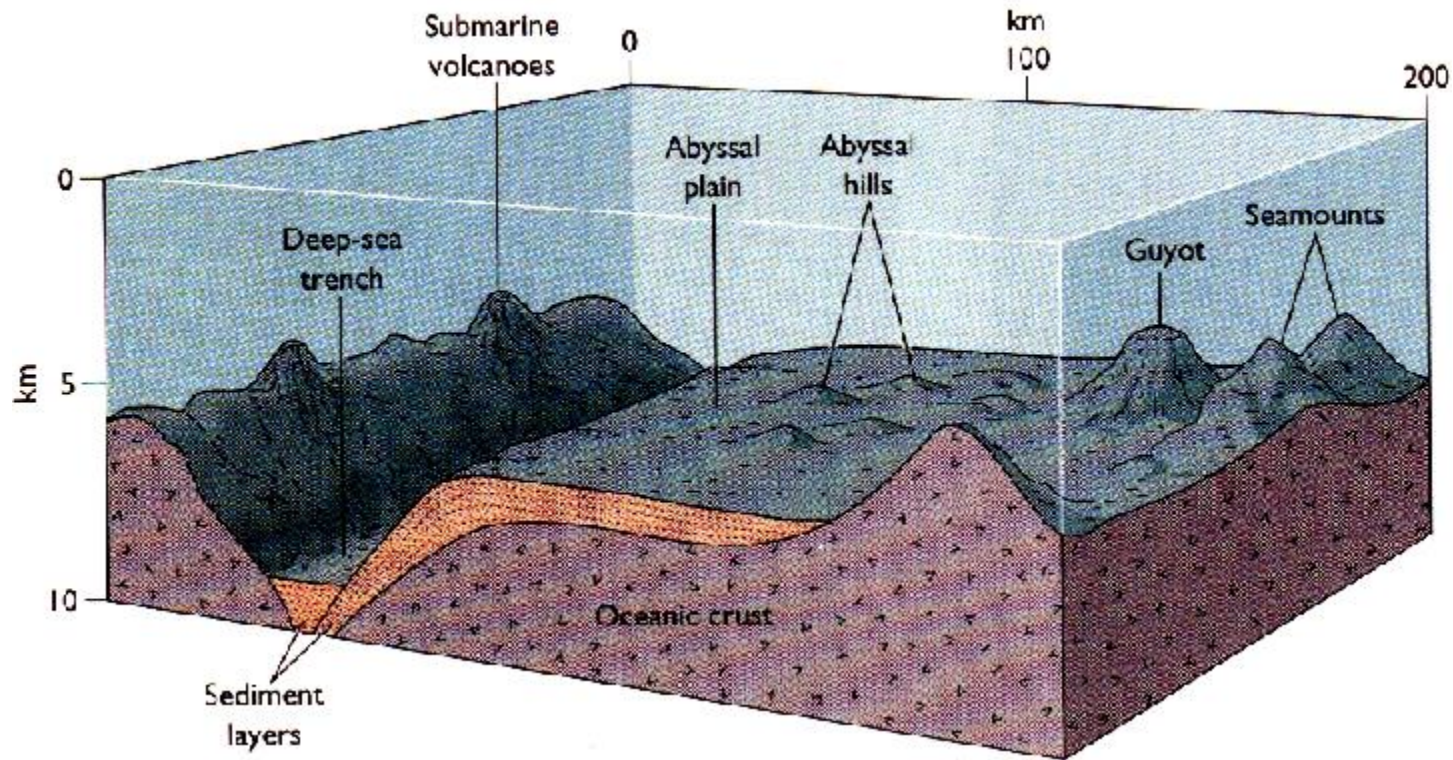
- The margins of continents are boundaries between continental crust and oceanic crust
- They generally consist of:
 - Continental shelf (most landward)
 - Continental slope
 - Continental rise (most seaward)





2. Deep ocean Basin (深海盆地)

- Covers the large area of the sea floor
- Very deep, from 3000 to 5000 m in depth
- Generally very flat, but still can have seamounts (were once islands, now several hundreds meters below sea surface because of lithosphere sunk and sea level rising)



3. Mid-ocean ridges and hydrothermal vents

- Plates pull apart at the center of ridges, form the central rift valley
- Heated sea water erupts from crusts and form the hydrothermal vents, water temperature could reach to 350 °C
- The minerals (mainly sulfide) in hot water solidified to form the black-smoke
- Could support a diversified and unique marine ecosystem

